



Maritime &
Coastguard
Agency

Guidance

MIN 542 (M+F) Amendment 3 life saving appliances – recognised distress signals and advertised alternatives to pyrotechnic flares

Published 12 April 2024

Contents

Summary

1. Introduction
2. Background
3. UK Policy on Marketed Alternatives to Pyrotechnic Flares
4. Advice for Mariners

More information



© Crown copyright 2024

This publication is licensed under the terms of the Open Government Licence v3.0 except where otherwise stated. To view this licence, visit nationalarchives.gov.uk/doc/open-government-licence/version/3 or write to the Information Policy Team, The National Archives, Kew, London TW9 4DU, or email: psi@nationalarchives.gov.uk.

Where we have identified any third party copyright information you will need to obtain permission from the copyright holders concerned.

This publication is available at <https://www.gov.uk/government/publications/min-542-mf-amendment-3-life-saving-appliances-recognised-distress-signals-and-advertised-alternatives-to-pyrotechnic-flares/min-542-mf-amendment-3-life-saving-appliances-recognised-distress-signals-and-advertised-alternatives-to-pyrotechnic-flares>

Summary

This Marine Information Note (MIN) provides the latest advice to mariners on products sold as distress signalling equipment to be used in an emergency. It highlights the risks of using non-SOLAS or non-pyrotechnic distress flares such as Electronic Visual Distress Signals (EVDS). This MIN (Amendment 3) expires 01 March 2027 and has been issued to incorporate the UK equipment type approval process.

1. Introduction

1.1 The International Regulations for Preventing Collisions at Sea, 1972 (COLREGS) apply to all vessels at sea and, by special application, to vessels on inland waters. Annex IV of these Regulations lists the signals which should be used if a vessel is in distress and needs immediate assistance. Distress alerting equipment to transmit these signals (e.g. pyrotechnics) must be carried on all seagoing commercial vessels and on most non-seagoing commercial vessels.

1.2 The specific distress alerting equipment which must be carried is prescribed in the carriage requirements for each type of vessel and nature of operation. In each case, a range of equipment is specified to transmit

an initial alert, long distance location signal, close range location signal or, in some cases, all three.

1.3 For pleasure vessels of 13.7m and over (Class XII) operating in Category C waters and seaward, three types of COLREGS distress alerting equipment must be carried to comply with the Class XII Exemption in MGN 599:

- a maritime radio capable of transmitting and receiving calls and messages appropriate for the area of operation – it is strongly recommended that the maritime radio provided has a digital selective calling (DSC) function;
- 4 red handheld flares and 2 orange smoke flares;
- another type of alerting equipment such as an EPIRB or a Personal Locator Beacon.

The manual raising and lowering of arms does not qualify as an appropriate means of signalling distress for meeting the carriage requirements of the Class XII pleasure vessel exemption in MGN 599.

1.4 For pleasure vessels of less than 13.7m, there are no specific carriage requirements, but the COLREGS list of recognised distress signals is still relevant and should be followed.

1.5 The International Convention for the Safety of Life at Sea, 1974 (SOLAS) describes the standards that distress alerting equipment should meet but there are also non-SOLAS distress signals on the market. The MCA has no mechanism for assessing the standard of these non-SOLAS signals, whereas SOLAS products sold in the UK are subject to a strict assessment and type approval process, and will be marked with the 'Red Ensign' Mark, although dependant upon the 'last placing onboard date' detailed in the MSN 1874, items marked with the Wheel Mark may still be accepted. The Maritime & Coastguard Agency (MCA) expects vessel operators to ensure that the distress alerting equipment they buy and carry is compliant with the applicable International Regulations in every respect.

2. Background

2.1 The MCA is aware of the development and marketing of hand-held non-pyrotechnic devices offered as alternatives to pyrotechnic flares; these will be referred to in this document as Electronic Visual Distress

Signals (EVDS), which covers an array of different devices. While these devices have a number of potential uses, from a practical and search and rescue perspective the light signal they produce is different to that produced by a pyrotechnic flare and may not be recognised as a distress signal. This could have fatal consequences.

3. UK Policy on Marketed Alternatives to Pyrotechnic Flares

3.1 To be effective, distress signals need to be internationally recognised. Not all EVDS provide a distress signal listed in COLREGS Annex IV (such as SOS), and the MCA knows of no EVDS device which is compliant with the SOLAS technical performance standards for distress alerting equipment. Consequently, the international carriage requirements do not recognise EVDS and the UK national carriage requirements have not been amended to formally recognise them either.

3.2 A change would therefore be needed to Annex IV of the COLREGS or to the SOLAS technical performance standards and to the applicable international/UK national carriage requirements in order to offer these devices full recognition as distress signals. The MCA sees no likely potential of this in the near future because objective evidence of a compelling need for change is yet to emerge.

3.3 The US Coast Guard has worked with the International Standards Organisation (ISO) and the Radio Technical Commission for Maritime Services (RTCM) to research the effectiveness of EVDS. The MCA supports this initiative, through monitoring progress, contributing views, and supporting this work in the appropriate international fora. The ultimate aim of the work is to consider the development of an appropriate technical performance standard for EVDS devices, if they are shown to be fit for purpose.

4. Advice for Mariners

4.1 For commercial vessels which are required to carry flares, any hand-held distress flares carried must meet the requirements of the UK marine equipment regulations demonstrated by the presence of the UK conformity mark for marine equipment - the Red Ensign, as detailed in MSN 1874 and MIN 590 Amendment 5- which incorporates UK and IMO

requirements. Dependant on the 'last placing onboard date' EU Marine Equipment Directive conformity marked items – denoted by the 'Wheel Mark'- may still be accepted onboard UK vessels, until the end of their serviceable life. The MCA is not aware of any EVDS product which meets the light intensity required by the IMO for distress flares. This means that if your commercial vessel is required to carry flares, they cannot be an EVDS product.

4.2 Where carriage of flares is not mandatory, the MCA nevertheless advises that EVDS should not currently be carried as a substitute for conventional pyrotechnic flares. This is because of the risk that EVDS may not be recognised internationally as a distress signal. However, for pleasure vessels, seagoing commercial vessels, and most non-seagoing commercial vessels, EVDS may be carried in addition to the required distress signalling equipment and used to identify location or transmit the S-O-S distress signal through a switch mechanism (just as a torch or other light-emitting device could be used). However, the limitations of EVDS devices should be recognised and anyone using them should be made aware of the type and quality of signal being generated.

More information

Ship Standards
Maritime and Coastguard Agency
Bay 2/23
Spring Place
105 Commercial Road
Southampton
SO15 1EG

Telephone: +44 (0)203 8172000

Email: marinetechnology@mcga.gov.uk

Website: [Maritime and Coastguard Agency - GOV.UK](https://www.gov.uk/government/organisations/maritime-and-coastguard-agency)
(<https://www.gov.uk/government/organisations/maritime-and-coastguard-agency>)

General enquiries: infoline@mcga.gov.uk

Please note that all addresses and telephone numbers are correct at time of publishing.



OGI

All content is available under the [Open Government Licence v3.0](#), except where otherwise stated



© Crown copyright